MIL-L-52696(ME)
3 August 1970
SUPERSEDING
PD-493A
8 December 1959

MILITARY SPECIFICATION

LAMINATING RESIN

1. SCOPE

1.1 Scope. This specification covers the requirements for an epoxy laminating resin and hardener for impregnating fiber glass cloth to form a rigid adhesive repair over a damaged and prepared fiber glass base.

1.2 Classification. The resin and hardener shall be furnished in the following sizes as specified:

Size A - Half-pint can filled as specified herein with a 2-ounce tube of hardener.

Size B - Pint can filled as specified herein with a 4-ounce tube of hardener.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.

Military

MIL-T-3689	- Tubes, Collapsible.

STANDARDS

Federal

FED. TEST METHOD STD. No. 406	- Plastics: Method of Testing.

FSC 8030

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The laminating resin shall be an epoxy polymer system and a prepared hardener. After thorough mixing, the resin and hardener shall cure at room temperature (60° to 85° F.) and shall form a rigid plastic with adhesion to fiber glass and silicate products.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish laminating resins for examination and test within the time frame specified to prove prior to

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starting production, that his production methods will produce laminating resins that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Epoxy resins. The epoxy resins shall be of a class of materials normally derived from bisphenol A and epichlorohydrin. The resins shall be supplied in formulation with pigments and silica gel (thixotrope), or adjusted in viscosity to not greater than 20,000 centipoises at 70° to 80° F. by the addition of silicates and pigments to a level not greater than 8 percent by weight. The epoxy resins shall be formulated to allow the mixed resin to permeate woven fiber glass reinforcement cloth, and to allow flow to a smooth, even surface without run, sag or creep, when applied to a prepared vertical solid fiber glass base. Size A shall be a half-pint container filled with 6 ounces of epoxy resins, and size B shall be a pint container filled with 12 ounces of epoxy resins.

3.4.1 Uncatalyzed resin. The uncatalyzed epoxy resin shall have the following physical properties:

Property	Values Min.	Max.	Test
Specific gravity	1.15	1.25	4.6.1
Viscosity, centipoises	7,000	20,000	4.6.2
Epoxy number	165	200	4.6.3

3.4.2 Catalyzed and cured resin. The resin, when prepared in accordance with 4.5.2, shall have the following property values:

Property	Values Min.	Max.	Test
Specific gravity	1.20	1.30	4.6.1.2
Tensil Strength, (psi)	9,000	----	4.6.4
Hardness, Rockwell "M"	100	120	4.6.5
Water absorption, (percent)	----	0.2	4.6.6
Gel time, hours	6	----	4.6.7
Flexural strength laminate psi dry	36,000	----	4.6.8.2
Flexural strength laminate psi wet	30,000	----	4.6.8.2

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3.4.3 Instruction sheet. The supplier shall provide each resin unit with an instruction sheet containing the following:

(a) Maximum usable storage life of the uncatalyzed resin.
(b) Any other pertinent information on storage and handling of the resin such as most suitable containers and precautions to be taken in handling the resin and the catalyst.

3.5 Hardener. The hardener or catalyst shall be a solution or a suspension of boron trifluorideamine catalysts or activating agents of polyamines, dibasic acid anhydrides which will promote chain growth and crosslinking in exothermic reaction when mixed with the epoxy resins at room temperature. The hardener shall be supplied in sufficient strength per unit of product to activate the epoxy resins to end-point. The hardener shall be placed as required in 2-ounce or 4-ounce polyethylene tubes conforming to MIL-T-3689, type II, class 4. Closure shall be made with leak-free screwcaps.

3.6 Identification marking. The cans and plastic tubes shall be identified in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.8).

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4.3 Preproduction inspection. The preproduction laminating resins shall be tested as specified in 4.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot and lot size. For the purpose of sampling, a lot shall consist of an integral quantity of the material produced in one manufacturing cycle. A manufacturing cycle shall be limited to 2,000 pounds of resin.

4.4.2 Sampling. For purposes of sampling, a lot or shipment in the form of "package lots" (cans, half-pint or pint) shall be limited to 2,000 pounds of resin. For each lot the number of individual packages to be selected at random shall be as follows:

Number of packages in the lot	Number of packages to be sampled
1 to 3	all
4 to 64	4
65 to 125	5
126 to 216	6
217 to 343	7
344 to 512	8

4.4.3 Tests. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.6.1 through 4.6.8. Failure of any test shall be cause for rejection of the lot from which the samples were taken.

4.5 Test conditions and preparation.

4.5.1 Test conditions. Standard test conditions shall be 73° F. plus or minus 2° F. and 50 plus or minus 4 percent relative humidity. Specimens shall be tested after being exposed to this temperature and humidity for 2 days.

4.5.2 Preparation of test specimens and peak exotherm and gel time tests. Weigh to the nearest gram approximately 700 grams of uncatalyzed resin and heat to 160° F. plus or minus 2° F. Add, while stirring, a stoichiometric amount of purified metaphenylenediamine having a melting point of 62.8° C. to the heated resin until solution is complete. The stoichiometric amount of metaphenylenediamine is derived from the epoxy number (see 4.6.3), and quantitatively utilized in accordance with the following:

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Epoxy number	Number of grams of metaphenylenediamine per 100 grams of uncatalyzed resin
165	16.39
167	16.19
169	16.00
171	15.81
173	15.63
175	15.45
177	15.28
179	15.11
181	14.94
183	14.78
185	14.62
187	14.46
189	14.31
191	14.16
193	14.01
195	13.87
197	13.72
199	13.59

Allow the solution to cool to 105° F., and pour into a 6 by 4 by 1 inch deep metal pan to a depth of 0.75 inch. Test this specimen for gel test (see 4.6.7). Allow the solution in the pan to cure for 24 hours at standard conditions. Reheat the cured casting in the pan until it has reached a temperature of 300° plus or minus 2° F. throughout the casting, then post cure for 4 hours at 300° F. After post curing, cool the cured resin to room temperature.

4.5.3 Test specimen forms. Test specimens for applicable cast resin test (see 4.6.1.2), shall be machined into desired forms from castings prepared as specified in 4.5.2.

4.6 Test procedures.

4.6.1 Specific gravity.

4.6.1.1 Uncatalyzed resin. The specific gravity of the uncatalyzed liquid resin shall be measured with an accurate balance. The average of three determinations shall be recorded. The specific gravity shall be expressed as the ratio of the weight of a given volume of the resin at 25° C. to that of an equal volume of water at the same temperature. Nonconformance to 3.4.1 shall constitute failure of this test.

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4.6.1.2 Cast resin. The specific gravity of the cast resin shall be determined by method 5011 of FED. TEST METHOD STD. No. 406, on three specimens. The average of the three determinations shall be reported. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.2 Viscosity. The viscosity in centipoises of the uncatalyzed resin shall be determined with an accurate viscosimeter. Nonconformance to 3.4.1 shall constitute failure of this test.

4.6.3 Epoxy number. The epoxy number of the uncatalyzed resin is the number of grams of resin containing one epoxy equivalent. It shall be determined as follows:

(a) In a 125 milliliter ground glass stoppered Erlenmeyer flask, dissolve 0.5000 grams of uncatalyzed resin in 25.00 milliliters of hydrochloric acid-p-dioxane reagent. This reagent shall be prepared by pipetting 4.00 milliliters of reagent grade concentrated hydrochloric acid into a 250 milliliter volumetric flask and filling to the mark with p-dioxane having a melting point of 11.5° C.
(b) The resin solution as prepared above shall be allowed to stand for 20 minutes, with occasional shaking.
(c) After standing, wash into a 250 milliliter beaker with 50 milliliters of distilled water followed with 40 milliliters of acetone.
(d) To the wash add 5 milliliters of reagent grade nitric acid and 0.2 grams of gelatin.
(e) With continuous stirring, the mixture in the beaker shall be titrated with 0.1000 normal silver nitrate solution, using a calomel (saturated) - silver electrode system (see figure 1) containing a 0.05 molar ammonium nitrate bridge, to detect the equivalence point potentiometrically. Reagent grade silver nitrate shall be used. The end point of the titration shall be determined by adding the silver nitrate dropwise near the equivalence point and choosing the volume where the greatest change in potential occurs.
(f) The epoxy number shall be calculated as follows:

$$\text{Epoxy No.} = \frac{\text{Weight of resin sample} \times 100}{0.0001\ (\text{ml of } 0.1\text{N } AgNO_3 \text{ for blank}) - (\text{ml of } AgNO_3 \text{ for HCl})}$$

Nonconformance to 3.4.1 shall constitute failure of this test.

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4.6.4 Tensile strength. The tensile strength of the catalyzed and cured resin shall be determined by FED. TEST METHOD STD. No. 406, method 1011 on test specimens constructed in accordance with figure 2. The rate of head travel for the tensile testing machine shall be not greater than 0.05 inch per minute while extensometer readings are made. The average of five determinations shall be recorded. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.5 Hardness. Rockwell "M" hardness on the cured specimens shall be determined as prescribed by FED. TEST METHOD STD. No. 406, method 1081. The average of four readings per casting shall be reported. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.6 Water absorption. Water absorption of the cured resin shall be determined as prescribed in FED. TEST METHOD STD. No. 406, method 7031. The average of three measurements shall be recorded. Nonconformace to 3.4.2 shall constitute failure of this test.

4.6.7 Gel time. The gel time shall be reported as the time required for the specimen prepared as specified in 4.5.2, to become stringy as determined by the lowering and raising of a tin wire into the reaction mixture. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.8 Flexural strength tests.

4.6.8.1 Specimens. Specimens shall be made from a laminate of three plys of fiber glass mat pre-impregnated with the epoxy resin mixed with the supplier's hardener and allowed to cure as specified in 4.5.2. The specimen shall be 0.125 inches thick plus or minus 0.010 inches. When curing is complete cut the laminate into strips 1/2 inch wide and 4 inches long.

4.6.8.2 Laminate properties. Using specimens prepared in 4.6.8.1, conduct the following tests prescribed in FED. TEST METHOD STD. No. 406 and as specified herein:

Test	Method No.
Flexural strength, Dry	1031
Flexural strength, Wet	1031

Wet specimens shall be immersed in boiling water for two hours and cooled to test conditions as specified in 4.5.1 prior to test. Test shall be made on soaked specimens. Nonconformance with 3.4.2 shall constitute failure of this test.

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4.7 Rejection and retest. Resin which has been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the contracting officer. Resin rejected after retest shall not be resubmitted without the specific approval of the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Materials and containers not as specified. Each incorrect material or container shall constitute one defect.
102.	Quantity of laminating resins and hardener not packaged in quantities as specified.
103.	Tubes of hardener not cushioned or placed in cells as specified.
104.	Intermediate packages not waterproof sealed as specified for level A.
105.	Strapping not zinc coated for level A packing.
106.	Weight limitation of the containers exceeded for packaging or packing.
107.	Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.4). The laminating resins shall be packaged in quantities as specified herein in cans conforming to PPP-C-96, type V, class 2. The hardener shall be packaged in quantities as specified herein in plastic tubes conforming to MIL-T-3689. The laminating resins unit packaged as specified shall be intermediate packaged in a box conforming to PPP-B-636, W5c or W6c. Size A laminating resins shall be packaged in units of 12 half-pint cans along with 12 2-ounce tubes of hardener. Size B laminating resins shall be packaged in units of six pint cans along with six 4-ounce tubes of hardener. The plastic tubes shall be cushioned or

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placed in cells to prevent damage by free movement within the box. After closure the boxes shall be waterproof sealed with tape as specified for slotted boxes in accordance with the appendix to the box specification.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. The laminating resins and hardener packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style I, in quantities not to exceed the weight limitation of the container. Strapping shall be zinc coated.

5.2.2 Level B. The intermediate containers (see 5.1) used to package the laminating resins shall serve as the shipping container. When specified, the intermediate containers shall be consolidated in a box conforming to PPP-B-636, V3c, or PPP-B-640, class 2. The weight of contents of the consolidated containers shall not exceed the weight limitation of the box specification. After closure, the boxes shall be waterproof sealed with tape as specified in the appendix to the applicable box specification.

5.2.3 Level C. The intermediate containers (see 5.1) used to package the laminating resins shall serve as the shipping container. At the option of the supplier, the intermediate packaging shall be consolidated to insure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or purchase order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The epoxy laminating resins are intended for use in repairing fiber glass boats by laminating impregnated fiber glass cloth over the prepared damaged areas, bruised surfaces or weakened hulls.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of containers required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2.1).
(d) Level of packing required (see 5.2).
(e) When intermediate packaging will be consolidated for level B (see 5.2.2).
(f) Any special marking required (see 5.3).

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6.3 Preproduction model. Any changes or deviations of production laminating resins from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish laminating resins conforming to this specification.

6.4 Level of packaging (see 5.1). Only a single method of packaging is suitable and adequate for this commodity. For purposes of the packaging/packing level marking, A/A, A/B, or A/C shall be used. Levels B and C packaging are not applicable to this item as the shipping container will also be used in field supply as a storage container. As such, it is susceptible to exposure to all environmental conditions in addition to rough handling. A lesser level of package would require repack prior to troop issue.

Custodian:

Army - ME

Review interest:

Army - MR

Preparing activity:

Army - ME

Project No. 8030-A019

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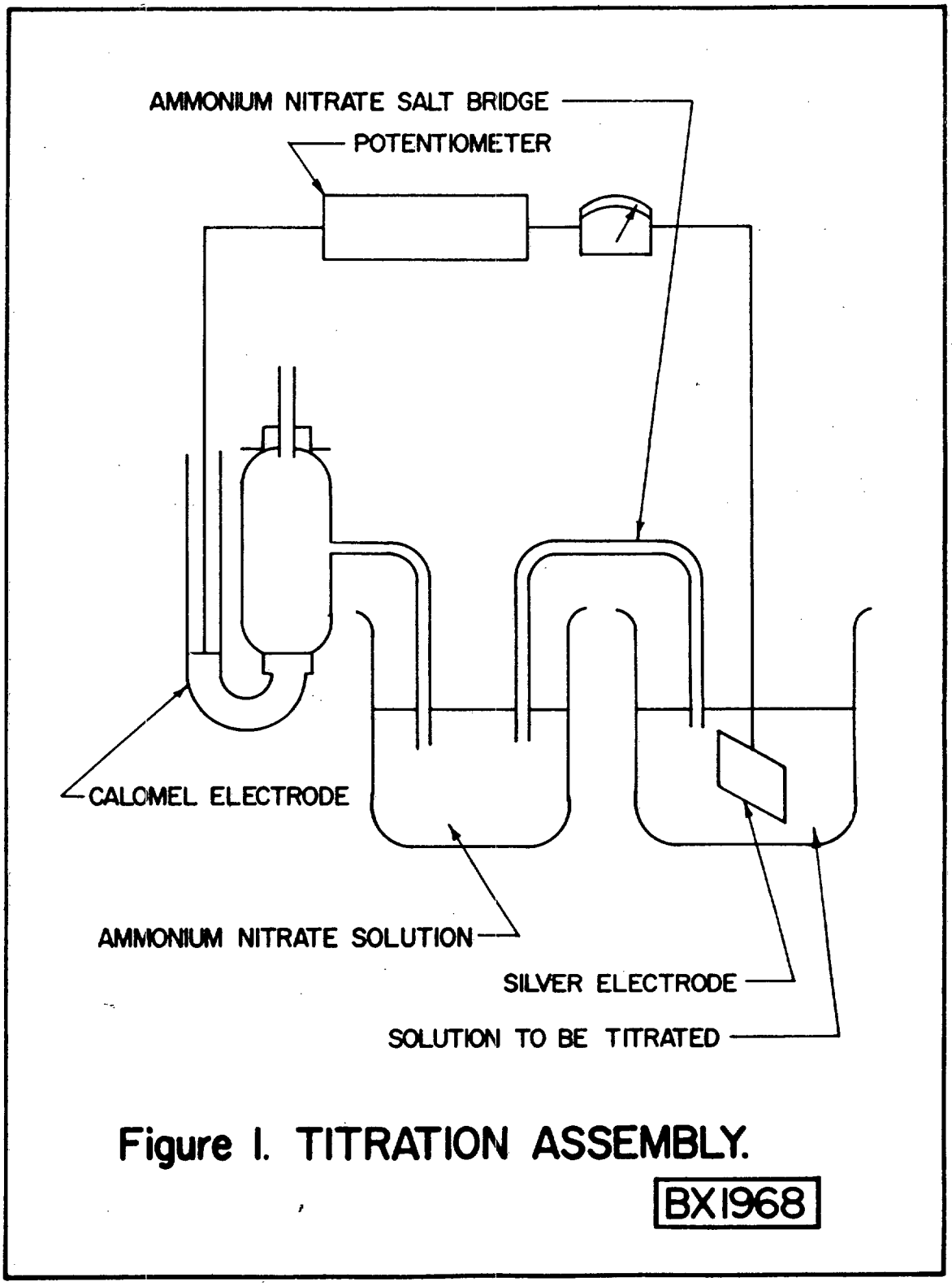


Figure I. TITRATION ASSEMBLY.

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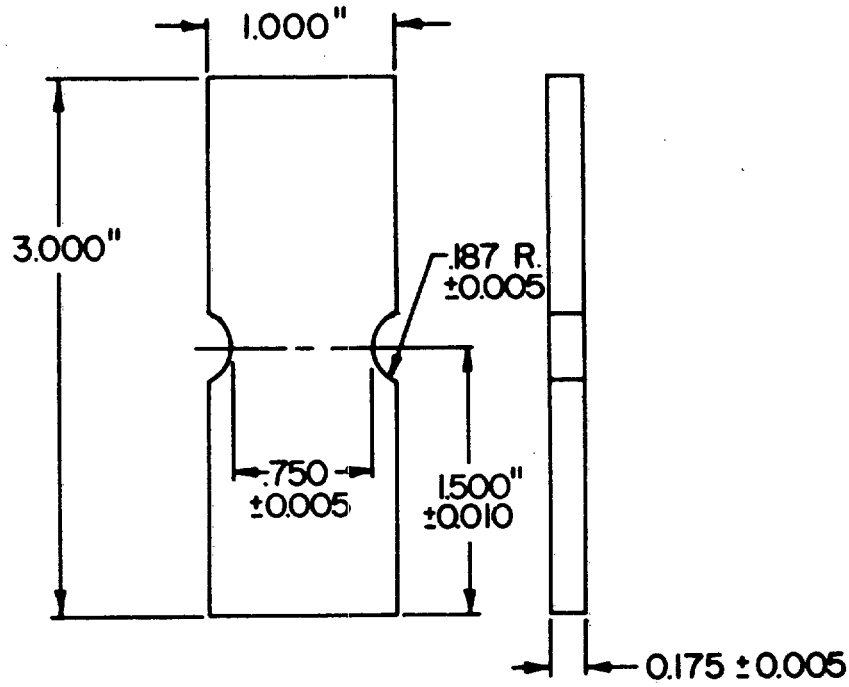


Figure 2. TENSILE TEST SPECIMEN

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-52696(ME) Laminating Resin

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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